Global Convergence for Low-Rank Mean-Field Neural Networks: Proof Outline

1 Setup: Low-Rank Mean-Field Equations

We consider a three-layer low-rank random feature network with the following structure:

- **First layer**: Random features $\varphi_1(\langle f_1(C_1),\, X\rangle)$ where $f_1(C_1)$ are frozen random vectors and C_1 indexes the first-layer channels.
- **Second layer**: Low-rank mixing matrix L of rank r, producing outputs

$$H_2(t, c_2; X, W) = \sum_{k=1}^{r} L_{c_2,k}\, m_k(t; X, W),$$

  where $m_k(t; X, W) = \mathbb{E}_{C_1}[w_1(t, C_1, k)\, \varphi_1(\langle f_1(C_1),\, X\rangle)]$ are the r channel-wise partial functions.
- **Third layer**: Trainable weights $w_2(t, C_2)$ producing the final output.

The mean-field ODE system consists of $r+1$ equations:

$$\partial_t w_1(t, c_1, k) = -\xi_1(t)\, \mathbb{E}_Z\left[d_L\big(Z; W(t)\big)\, \varphi_1(\langle f_1(c_1),\, X\rangle)\; B_k(t; X)\right], \tag{1}$$

$$\partial_t w_2(t, c_2) = -\xi_2(t)\, \mathbb{E}_Z\Big[d_L\big(Z; W(t)\big)\, \varphi_2\Big(H_2(t, c_2; X, W(t))\Big)\Big], \tag{2}$$

where $d_L(Z; W) = \partial_2 \mathcal{L}(Y, \hat{y}(X; W))$ is the loss derivative, and

$$B_k(t; X) = \mathbb{E}_{C_2}\Big[L_{C_2,k}\, \varphi_2'\Big(H_2(t, C_2; X, W(t))\Big)\, w_2(t, C_2)\Big]$$

is the channel-k backpropagated signal.

The model output is:

$$\hat{y}(X; W(t)) = \mathbb{E}_{C_2}\big[w_2(t, C_2)\, \varphi_2\big(H_2(t, C_2; X, W(t))\big)\big].$$

2 Assumptions for Global Convergence

Assumption 1 (Initialization)**.** *The initial functions $w_1^0 : \Omega_1 \times \{1, \ldots, r\} \to \mathbb{R}$ and $w_2^0 : \Omega_2 \to \mathbb{R}$ satisfy:*

$$\sup_{m\geq 1} \frac{1}{\sqrt{m}} \max_{1\leq k\leq r} \mathbb{E}_{C_1}\left[|w_1^0(C_1, k)|^m\right]^{1/m} \leq K,$$

$$\sup_{m\geq 1} \frac{1}{\sqrt{m}} \mathbb{E}_{C_2}\left[|w_2^0(C_2)|^m\right]^{1/m} \leq K.$$

Assumption 2 (Diversity). *The support of ρ^1 (the measure on Ω_1) is $\mathbb{R}^d$ (or dense in $\mathbb{R}^d$). This ensures that the random features $\varphi_1(\langle f_1(C_1), \cdot\rangle)$ have dense span in $L^2(\mathcal{P}_X)$.*

Assumption 3 (Regularity). *There exists $K \geq 1$ such that:*

- *ξ_1, ξ_2 are bounded on compact intervals and piecewise continuous.*
- *$\varphi_1, \varphi_2, \varphi_2'$ are K-bounded and K-Lipschitz; moreover φ_2' is non-vanishing (strictly non-zero everywhere).*
- *$\partial_2\mathcal{L}(y, \cdot)$ is K-bounded and K-Lipschitz for all y in the support of $\mathcal{P}$.*
- *X is K-sub-Gaussian and $\|f_1(c_1)\| \leq K$ for all c_1.*
- *The mixing kernel satisfies $\sup_{c_2,k} |L_{c_2,k}| \leq K$ and $k \mapsto L_{c_2,k}$ is measurable for each c_2.*
- *$|X| \leq K$ with probability 1.*

Assumption 4 (Convergence). *There exist functions $\bar{w}_1 : \Omega_1 \times \{1, \ldots, r\} \to \mathbb{R}$ and $\bar{w}_2 : \Omega_2 \to \mathbb{R}$ such that as $t \to \infty$, there exists a coupling π_t of $\rho^1 \times \rho^2$ and itself such that:*

$$\int (1 + |\bar{w}_2(c_2)|)|\bar{w}_2(c_2)| \max_{1\leq k\leq r} |\bar{w}_1(c_1, k)|\, |w_1^*(t, c_1', k) - \bar{w}_1(c_1, k)|\, d\pi_t(c_1, c_2, c_1', c_2') \to 0, \tag{3}$$

$$\int (1 + |\bar{w}_2(c_2)|)|\bar{w}_2(c_2)|\, |w_2^*(t, c_2') - \bar{w}_2(c_2)|\, d\pi_t(c_1, c_2, c_1', c_2') \to 0, \tag{4}$$

where $W^*(t) = (w_1^*(t, \cdot, \cdot), w_2^*(t, \cdot))$ is the solution to the mean-field ODEs (1)–(2).

Remark 1. *The convergence assumption (3)–(4) is adapted from the full-width case. The key difference is that we have r channels for w_1, so we take the maximum over $k \in \{1, \ldots, r\}$ in (3). The low-rank structure enters through the factor $\max_k |\bar{w}_1(c_1, k)|$ which is controlled by the ψ_2 bounds.*

3 Well-Posedness: Existence and Uniqueness of Solutions

We now provide a rigorous proof of existence and uniqueness for the mean-field ODE system (1)–(2). The proof follows a Picard iteration argument, adapted to the low-rank structure.

3.1 Norms and Spaces for Low-Rank Mean-Field Parameters

We equip the neural network parameters and their mean-field limits with several norms. For the mean-field parameters $W = (w_1, w_2)$, we define:

$$\begin{aligned}
w_{1t} &= \max_{1\leq k\leq r} \mathbb{E}_{C_1}\left[\sup_{s\leq t} |w_1(s, C_1, k)|^{50}\right]^{1/50}, \\
w_{2t} &= \mathbb{E}_{C_2}\left[\sup_{s\leq t} |w_2(s, C_2)|^{50}\right]^{1/50}, \\
W_t &= \max\left(w_{1t}, w_{2t}\right).
\end{aligned}$$

We also define the L^2-type norms:

$$\|w_1\|_t = \max_{1\le k\le r} \mathbb{E}_{C_1}\left[\sup_{s\le t} |w_1(s,C_1,k)|^2\right]^{1/2},$$
$$\|w_2\|_t = \mathbb{E}_{C_2}\left[\sup_{s\le t} |w_2(s,C_2)|^2\right]^{1/2},$$
$$\|W\|_t = \max\big(\|w_1\|_t, \|w_2\|_t\big).$$

Note that $\|W\|_t$ defines a norm on the space of mean-field parameters. For two sets of mean-field parameters W and W', we define the distance:

$$\|W - W'\|_t = \max\big(\|w_1 - w_1'\|_t, \|w_2 - w_2'\|_t\big), \tag{5}$$
$$\|w_1 - w_1'\|_t = \max_{1\le k\le r} \mathbb{E}_{C_1}\left[\sup_{s\le t} |w_1(s,C_1,k) - w_1'(s,C_1,k)|^2\right]^{1/2},$$
$$\|w_2 - w_2'\|_t = \mathbb{E}_{C_2}\left[\sup_{s\le t} |w_2(s,C_2) - w_2'(s,C_2)|^2\right]^{1/2}.$$

For convenience, we define the random variables:

$$\mathsf{max}_t^w(W) = \max\left(\max_{1\le k\le r}\sup_{s\le t}|w_1(s,C_1,k)|, \sup_{s\le t}|w_2(s,C_2)|\right),$$
$$\mathsf{max}_t^{w_1}(W) = \max_{1\le k\le r}\sup_{s\le t}|w_1(s,C_1,k)|.$$

3.2 Sub-Gaussian Norms

For the well-posedness argument, we consider the following sub-Gaussian (ψ_2) norms:

$$[\![w]\!]_{1\psi,t} = \sqrt{50}\,\sup_{m\ge 1}\frac{1}{\sqrt{m}}\max_{1\le k\le r}\mathbb{E}_{C_1}\left[\sup_{s\le t}|w_1(s,C_1,k)|^m\right]^{1/m},$$
$$[\![w]\!]_{2\psi,t} = \sqrt{50}\,\sup_{m\ge 1}\frac{1}{\sqrt{m}}\mathbb{E}_{C_2}\left[\sup_{s\le t}|w_2(s,C_2)|^m\right]^{1/m},$$
$$[\![W]\!]_{\psi,t} = \max\big([\![w]\!]_{1\psi,t}, [\![w]\!]_{2\psi,t}\big).$$

The factor $\sqrt{50}$ ensures that $[\![w]\!]_{i\psi,t} \ge w_{it}$ and hence $[\![W]\!]_{\psi,t} \ge W_t$.

3.3 Solution Operator and Fixed Point Formulation

Denote by $\mathfrak{W}_T$ the space of mean-field parameters W such that $\|W\|_T < \infty$. Given a terminal time $T \ge 0$ and an initialization $W(0)$, we define the mapping F that associates $W' \in \mathfrak{W}_T$ with:

$$F(W')(t) = \big(F_1(W')(t,\cdot,\cdot), F_2(W')(t,\cdot)\big),$$

where

$$F_1(W')(t,c_1,k) = w_1(0,c_1,k) - \int_0^t \xi_1(s)\,\mathbb{E}_Z\big[d_L\big(Z;W'(s)\big)\,\varphi_1(\langle f_1(c_1),\,X\rangle)\;B_k(s;X,W'(s))\big]\,ds,$$
$$F_2(W')(t,c_2) = w_2(0,c_2) - \int_0^t \xi_2(s)\,\mathbb{E}_Z\Big[d_L\big(Z;W'(s)\big)\,\varphi_2\Big(H_2(s,c_2;X,W'(s))\Big)\Big]\,ds,$$

and $B_k(s; X, W') = \mathbb{E}_{C_2}[L_{C_2,k}\, \varphi_2'(H_2(s, C_2; X, W'))\, w_2'(s, C_2)]$.

Observe that at initialization $F(W')(0, \cdot, \cdot) = W(0)$, whereas the quantities in the time integrals are computed with respect to W'. A solution W to the mean-field ODEs on $[0, T]$ is an element of $\mathfrak{W}_T$ satisfying $F(W) = W$. We say that W is a solution on $t \in [0, \infty)$ if its restriction to $[0, T]$ is a solution on $[0, T]$ for all $T > 0$.

Theorem 1 (Existence and Uniqueness for Low-Rank Mean-Field ODEs)**.** *Assume that the initialization $W(0)$ of the mean-field ODEs satisfies $[\![W]\!]_{\psi,0} \le K$ for some constant $K > 0$. Then under Assumptions 3, 1, and* **??**, *there exists a unique solution to the mean-field ODEs* (1)–(2) *on $t \in [0, \infty)$.*

3.4 A Priori Bounds

Lemma 1 (A priori bounds for low-rank mean-field ODEs)**.** *Under Assumptions 3 and* **??**, *given an initialization $W(0)$, a solution W to the mean-field ODEs, if it exists, must satisfy that for any $t \in [0, \infty)$:*

$$W_t \le K_0(t),$$

where $K_0(t)$ is a non-decreasing function of the form

$$K_0(t) = K^\kappa (1 + t^\kappa)(1 + W_0^\kappa),$$

for some constant $\kappa > 0$ depending on K and r (the rank), but independent of the width N.

A similar result holds for the ψ_2 norm. Under the same assumptions, for any $t \in [0, \infty)$, there exists $K_0(t) \ge 1$ of the form

$$K_0(t) = K^\kappa (1 + t^\kappa)(1 + [\![W]\!]_{\psi,0}^\kappa),$$

such that a solution W, if it exists, must satisfy $W_t \le [\![W]\!]_{\psi,t} \le K_0(t)$. Furthermore, by assuming $[\![W]\!]_{\psi,0} < \infty$, for any $B \ge 0$:

$$\mathbb{P}\big(\mathsf{max}_t^w(W) \ge K_0(t) B\big) \le C e^{-cB^2},$$

for some universal constants $C, c > 0$.

Proof sketch. The proof follows the standard mean-field argument, with the key modification that the low-rank structure enters through the factor $\|L\|_{\infty,1} \le rK$ in forward propagation bounds.

For w_2, the drift is $-\xi_2 \mathbb{E}[d_L\, \varphi_2(H_2)]$, which is bounded by K times the bound on H_2. For H_2, we have:

$$|H_2(t, c_2; X, W)| \le \sum_{k=1}^r |L_{c_2,k}|\, |m_k(t; X, W)| \le \|L\|_{\infty,1} \max_k |m_k(t; X, W)| \le rK \max_k \mathbb{E}_{C_1}[|w_1(t, C_1, k)|].$$

This gives a bound of the form:

$$w_{2t} \le K(1 + rK\, w_{1t}) \le K(1 + rK\, W_t).$$

For w_1, the drift involves B_k, which is bounded by:

$$|B_k(t; X, W)| \le K\, \mathbb{E}_{C_2}[|L_{C_2,k}|\, |w_2(t, C_2)|] \le K\, \|L\|_{\infty,1}\, w_{2t} \le rK^2\, w_{2t}.$$

Combining these bounds and using Grönwall-type arguments yields the polynomial bound $K_0(t)$. The sub-Gaussian tail bound follows from the ψ_2 control and a union bound over the r channels. □

3.5 Invariance and Contraction Properties

The a priori bounds lead us to consider the following spaces, given an initialization $W(0)$ and an arbitrary terminal time $T > 0$:

- The space $\mathcal{W}_T$ of mean-field parameters $W' = \{W'(t)\}_{t\le T} = \{w_1'(t,\cdot,\cdot), w_2'(t,\cdot)\}_{t\le T}$ such that $W_T' \le K_0(T)$.
- The space $\mathcal{W}_T^0 \subset \mathcal{W}_T$ of mean-field parameters $W' \in \mathcal{W}_T$ such that:

$$[\![W]\!]'_{\psi,T} \le K_0(T),$$
$$\mathbb{P}\big(\mathsf{max}_T^w(W') \ge K_0(T)B\big) \le Ce^{-cB^2} \quad \forall B \ge 0,$$

  and $W'(0) = W(0)$ (so all elements in $\mathcal{W}_T^0$ share the same initialization).

We equip these spaces with the metric $(W', W'') \mapsto \|W' - W''\|_T$. By Lemma **??**, any solution W to the mean-field ODEs, if it exists, must belong to $\mathcal{W}_T^0$.

Lemma 2 (Invariance of the solution operator)**.** *Under Assumptions 3 and* **??***, for any $W' \in \mathcal{W}_T^0$, we have $F(W') \in \mathcal{W}_T^0$.*

Proof sketch. The integral form defining F applies bounded/Lipschitz drifts to W', so the same moment and tail bounds propagate with constants controlled by $K_0(T)$. The low-rank structure enters through the factor rK in the bounds, but this is absorbed into $K_0(T)$. □

Lemma 3 (Difference estimate for the solution operator)**.** *For a given $B \ge 0$, consider two collections of mean-field parameters $W', W'' \in \mathcal{W}_T$ such that:*

$$\mathbb{P}\big(\mathsf{max}_T^w(W') \ge K_0(T)B\big) \le Ce^{-cB^2},$$
$$\mathbb{P}\big(\mathsf{max}_T^w(W'') \ge K_0(T)B\big) \le Ce^{-cB^2}.$$

*Under Assumptions 3–***??***, for any $t \le T$:*

$$\|F(W') - F(W'')\|_t \le (KK_0(T))^4 \int_0^t \Big((1+B)\|W' - W''\|_s + e^{-cB^2/2}\Big) ds,$$

where the constant depends on r through $\|L\|_{\infty,1} \le rK$, but the structure of the estimate remains the same as in the full-width case.

Proof sketch. The key steps are:

1. **Forward Lipschitz**: Using Lemma 1, we have:

$$\mathbb{E}\left[\sup_{s\le t}\sup_{c_2} |H_2(s,c_2;X,W') - H_2(s,c_2;X,W'')|^2\right]^{1/2} \le K\|L\|_{\infty,1}\, \|w_1'-w_1''\|_t \le rK^2\, \|W'-W''\|_t.$$

2. **Backward Lipschitz**: On the good event $\{\mathsf{max}_T^w \le K_0(T)B\}$, the drift differences are Lipschitz with constant $(1+B)K_0(T)^2$. The bad event contributes an exponentially small term $e^{-cB^2/2}$.

3. **Integration**: Combining these bounds and integrating in time yields the estimate. The factor $(KK_0(T))^4$ comes from the product of Lipschitz constants, with the r factor from low-rank structure absorbed into K.

□

3.6 Proof of Existence and Uniqueness

Proof of Theorem **??**. We perform a Picard-type iteration argument. Consider an arbitrary finite $T \geq 0$ and $W', W'' \in \mathcal{W}_T^0$. From Lemma **??**:

$$\|F(W') - F(W'')\|_t \leq (KK_0(T))^4 \left((1+B) \int_0^t \|W' - W''\|_s \, ds + Te^{-cB^2/2} \right) \equiv k_1(1+B) \int_0^t \|W' - W''\|_s \, ds + k_2 e^{-k_3}$$

for any $B > 0$, where $k_1 = (KK_0(T))^4$, $k_2 = (KK_0(T))^4 T$, and $k_3 = c/2$.

By Lemma **??**, F maps $\mathcal{W}_T^0$ to $\mathcal{W}_T^0$. We can iterate this inequality to obtain:

$$\begin{aligned}
\|F^{(m)}(W') - F^{(m)}(W'')\|_T &\leq k_1(1+B) \int_0^T \|F^{(m-1)}(W') - F^{(m-1)}(W'')\|_{T_2} \, dT_2 + k_2 e^{-k_3 B^2} \\
&\leq k_1^2 (1+B)^2 \int_0^T \int_0^{T_2} \|F^{(m-2)}(W') - F^{(m-2)}(W'')\|_{T_3} \, \mathbb{I}(T_2 \leq T) \, dT_3 \, dT_2 \\
&\quad + k_2 \sum_{\ell=1}^{2} \frac{(Tk_1(1+B))^{\ell-1}}{\ell!} e^{-k_3 B^2} \\
&\cdots \\
&\leq \frac{1}{m!} T^m k_1^m (1+B)^m \|W' - W''\|_T + k_2 e^{Tk_1(1+B) - k_3 B^2}.
\end{aligned}$$

Choosing $B = \sqrt{m}$:

$$\|F^{(m)}(W') - F^{(m)}(W'')\|_T \leq \frac{1}{m!} T^m k_1^m (1+\sqrt{m})^m \|W' - W''\|_T + k_2 e^{Tk_1(1+\sqrt{m}) - k_3 m}.$$

Note that since $W_0 < \infty$, $K_0(T)$ and hence k_1, k_2 are finite for finite T. By substituting $W'' = F(W')$, we obtain:

$$\sum_{m=1}^{\infty} \|F^{(m+1)}(W') - F^{(m)}(W')\|_T = \sum_{m=1}^{\infty} \|F^{(m)}(W'') - F^{(m)}(W')\|_T < \infty.$$

Hence as $m \to \infty$, $F^{(m)}(W')$ converges in $\|\cdot\|_T$ to a limit $W \in \mathfrak{W}_T$, which is a fixed point of F. By Lemma **??**, W belongs to $\mathcal{W}_T^0$.

The uniqueness of the fixed point follows from the above estimate: if W' and W'' are fixed points of F, then they are both in $\mathcal{W}_T^0$, and:

$$\|W' - W''\|_T = \|F^{(m)}(W') - F^{(m)}(W'')\|_T \leq \frac{1}{m!} T^m k_1^m (1+\sqrt{m})^m \|W' - W''\|_T + k_2 e^{Tk_1(1+\sqrt{m}) - k_3 m}.$$

Taking m arbitrarily large, the right-hand side tends to 0, proving uniqueness. Since T is arbitrary, we have existence and uniqueness of the solution on $[0, \infty)$. $\square$

Remark 2 (Key differences from full-width case)**.** *The proof structure is identical to the full-width case, with the following adaptations:*

- *The norm $\|w_1\|_t$ takes the maximum over r channels:* $\max_{1 \leq k \leq r} \mathbb{E}_{C_1}[\sup_{s \leq t} |w_1(s, C_1, k)|^2]^{1/2}$.
- *The forward propagation bound includes a factor $\|L\|_{\infty,1} \leq rK$, but this is absorbed into the constants $K_0(T)$.*
- *The contraction constant becomes $(KK_0(T))^4$ instead of $(KK_0(T))^{2L+2}$ for L-layer networks, reflecting the simpler two-layer structure (with low-rank mixing).*
- *The sub-Gaussian tail bound involves a union bound over r channels, but the exponential decay rate remains the same.*

4 Key Technical Lemma: Bi-Lipschitz Property of H_2

Lemma 4 (Bi-Lipschitz property of H_2 with low-rank structure)**.** *For any $W' = (w_1', w_2')$ and $W'' = (w_1'', w_2'')$ satisfying the regularity assumptions, we have:*

$$\big|H_2(t, c_2; X, W') - H_2(t, c_2; X, W'')\big| \leq K \, \|L\|_{\infty,1} \max_{1\leq k\leq r} \mathbb{E}_{C_1}\big[|w_1'(t, C_1, k) - w_1''(t, C_1, k)|\big],$$

where $\|L\|_{\infty,1} \equiv \sup_{c_2} \sum_{k=1}^r |L_{c_2,k}| \leq rK$ under the entrywise bound.

Moreover, if φ_2 is ReLU or a Lipschitz activation, then:

$$\big|\varphi_2(H_2(t, c_2; X, W')) - \varphi_2(H_2(t, c_2; X, W''))\big| \leq K \, \|L\|_{\infty,1} \max_{1\leq k\leq r} \mathbb{E}_{C_1}\big[|w_1'(t, C_1, k) - w_1''(t, C_1, k)|\big].$$

Proof sketch. By definition:

$$\begin{aligned} H_2(t, c_2; X, W') - H_2(t, c_2; X, W'') &= \sum_{k=1}^r L_{c_2,k}\big(m_k(t; X, W') - m_k(t; X, W'')\big) \\ &= \sum_{k=1}^r L_{c_2,k}\mathbb{E}_{C_1}\big[(w_1'(t, C_1, k) - w_1''(t, C_1, k))\, \varphi_1(\langle f_1(C_1),\, X\rangle)\big]. \end{aligned}$$

Taking absolute values and using boundedness of φ_1:

$$\begin{aligned} \big|H_2(t, c_2; X, W') - H_2(t, c_2; X, W'')\big| &\leq \sum_{k=1}^r |L_{c_2,k}|\, \mathbb{E}_{C_1}\big[|w_1'(t, C_1, k) - w_1''(t, C_1, k)|\big]\, |\varphi_1(\langle f_1(C_1),\, X\rangle)| \\ &\leq K \sum_{k=1}^r |L_{c_2,k}| \max_{1\leq k\leq r} \mathbb{E}_{C_1}\big[|w_1'(t, C_1, k) - w_1''(t, C_1, k)|\big] \\ &\leq K\, \|L\|_{\infty,1} \max_{1\leq k\leq r} \mathbb{E}_{C_1}\big[|w_1'(t, C_1, k) - w_1''(t, C_1, k)|\big]. \end{aligned}$$

The second part follows from the Lipschitz property of φ_2. □

5 Proof Outline: Global Convergence

Theorem 2 (Global Convergence for Low-Rank Networks)**.** *Under Assumptions 1–4, if the loss $\mathcal{L}$ is convex in its second variable, then the limit point $(\bar{w}_1, \bar{w}_2)$ is a global minimizer of the population loss:*

$$\mathscr{L}(\bar{w}_1, \bar{w}_2) = \mathbb{E}_Z\big[\mathcal{L}\big(Y, \hat{y}(X; \bar{w}_1, \bar{w}_2)\big)\big].$$

5.1 Detailed Proof (Adapted from Full-Width Case)

We now provide the detailed proof, adapting the argument from the full-width three-layer case to our low-rank setting. The key simplification is that we do not train ρ^1 (the first layer uses frozen random features), so we avoid the algebraic topology arguments needed to show full support of $\mathrm{Law}(w_1^*(t, C_1))$.

Step 1: Convergence of the backpropagated signal

By the convergence assumption (Assumption 4), we have that for any $\epsilon > 0$, there exists $T(\epsilon)$ such that for all $t \geq T(\epsilon)$ and almost surely:

$$\epsilon \geq \big|\mathbb{E}_Z\big[d_L\big(Z; W^*(t)\big)\, \varphi_1(\langle f_1(C_1),\, X\rangle)\, B_k^*(t; X)\big]\big| = \Big|\big\langle \mathbb{E}_Z\big[d_L\big(Z; W^*(t)\big)\, B_k^*(t; X)\big|X = x\big]\,, \varphi_1(\langle f_1(C_1),\, x\rangle)\big\rangle_{L^2(\mathcal{P}_X)}\Big|$$

where $B_k^*(t;X) = \mathbb{E}_{C_2}[L_{C_2,k}\,\varphi_2'(H_2^*(t,C_2;X,W^*(t)))\,w_2^*(t,C_2)]$.

Since $f_1(C_1)$ are frozen random features with full support in $\mathbb{R}^d$ (by Assumption 2), and we have r channels, the set $\{\varphi_1(\langle f_1(c_1),\,\cdot\rangle) : c_1 \in \Omega_1, k \in \{1,\dots,r\}\}$ has dense span in $L^2(\mathcal{P}_X)$. We thus obtain that for u in a dense subset of $\mathbb{R}^d$:

$$\text{ess-sup}\left|\left\langle \mathbb{E}_Z\left[d_L\big(Z;W^*(t)\big)\,B_k^*(t;X)\middle|X=x\right],\varphi_1(\langle u,\,x\rangle)\right\rangle_{L^2(\mathcal{P}_X)}\right| \le \epsilon.$$

By continuity of $u \mapsto \varphi_1(\langle u,\,\cdot\rangle)$ in $L^2(\mathcal{P}_X)$, we extend the above to all $u \in \mathbb{R}^d$.

Step 2: Coupling argument with low-rank structure

Recall the couplings π_t in Assumption 4. Since φ_1 is bounded, we have:

$$\begin{aligned}&\mathbb{E}_{(C_1,C_2,C_1',C_2')\sim\pi_t}\left[\left|\left\langle \mathbb{E}_Z\left[d_L\big(Z;W^*(t)\big)\,B_k^*(t;X) - d_L\big(Z;\bar{W}\big)\,\bar{B}_k(X)\middle|X=x\right],\varphi_1(\langle u,\,x\rangle)\right\rangle_{L^2(\mathcal{P}_X)}\right|\right]\\ &\quad\le K\,\mathbb{E}_{\pi_t}\left[\left|d_L\big(Z;W^*(t)\big)\,B_k^*(t;X) - d_L\big(Z;\bar{W}\big)\,\bar{B}_k(X)\right|\right],\end{aligned}$$

where $\bar{W} = (\bar{w}_1,\bar{w}_2)$ and $\bar{B}_k(X) = \mathbb{E}_{C_2}[L_{C_2,k}\,\varphi_2'(\bar{H}_2(C_2;X,\bar{W}))\,\bar{w}_2(C_2)]$ with $\bar{H}_2(c_2;X,\bar{W}) = \sum_{k=1}^r L_{c_2,k}\,\bar{m}_k(X,\bar{W})$ and $\bar{m}_k(X,\bar{W}) = \mathbb{E}_{C_1}[\bar{w}_1(C_1,k)\,\varphi_1(\langle f_1(C_1),\,X\rangle)]$.

Step 3: Key bound using bi-Lipschitz property (adapted from full-width case)

Using the regularity assumption and Lemma 1, similar to the calculation in the full-width case, we bound:

$$\begin{aligned}&\mathbb{E}_{\pi_t}\left[\left|d_L\big(Z;W^*(t)\big)\,B_k^*(t;X) - d_L\big(Z;\bar{W}\big)\,\bar{B}_k(X)\right|\right]\\ &\quad\le K\,\mathbb{E}_{\pi_t}\Big[(1+|\bar{w}_2(C_2)|)\Big(|w_2^*(t,C_2') - \bar{w}_2(C_2)|\\ &\qquad + |\bar{w}_2(C_2)|\,\max_{1\le k\le r}|w_1^*(t,C_1',k) - \bar{w}_1(C_1,k)|\Big)\Big],\end{aligned}$$

where the last step uses:

- The Lipschitz property of $\partial_2\mathcal{L}$ and φ_2'
- The bi-Lipschitz bound from Lemma 1 for H_2 differences: $|H_2^*(t) - H_2(\bar{W})| \le K\|L\|_{\infty,1}\max_k|w_1^*(t) - \bar{w}_1|$
- The structure $B_k = \mathbb{E}_{C_2}[L_{C_2,k}\,\varphi_2'(H_2)\,w_2]$, which gives:

$$\begin{aligned}|B_k^*(t) - B_k(\bar{W})| &\le K\mathbb{E}_{C_2}\left[|L_{C_2,k}|\big(|\varphi_2'(H_2^*(t)) - \varphi_2'(\bar{H}_2)|\,|w_2^*(t)| + |\varphi_2'(\bar{H}_2)|\,|w_2^*(t) - \bar{w}_2|\big)\right]\\ &\le K(1+|\bar{w}_2|)\Big(|w_2^*(t) - \bar{w}_2| + |\bar{w}_2|\,\max_k|w_1^*(t) - \bar{w}_1|\Big)\end{aligned}$$

By Assumption 4, the right-hand side converges to 0 as $t\to\infty$.

Step 4: Orthogonality and optimality

We thus obtain that for all $u \in \mathbb{R}^d$:

$$\mathbb{E}_{C_2}\left[\left|\left\langle \mathbb{E}_Z\left[d_L\big(Z;\bar{W}\big)\,\bar{B}_k(X)\middle|X=x\right],\varphi_1(\langle u,\,x\rangle)\right\rangle_{L^2(\mathcal{P}_X)}\right|\right] = 0,$$

which yields that for all $u \in \mathbb{R}^d$ and almost surely:

$$\left|\left\langle \mathbb{E}_Z\left[d_L\big(Z;\bar{W}\big)\,\bar{B}_k(X)\middle|X=x\right],\varphi_1(\langle u,\,x\rangle)\right\rangle_{L^2(\mathcal{P}_X)}\right| = 0.$$

Since $\{\varphi_1(\langle u,\,\cdot\rangle) : u \in \mathbb{R}^d\}$ has dense span in $L^2(\mathcal{P}_X)$ (by Assumption 2), we have:

$$\mathbb{E}_Z\left[d_L\big(Z;\bar{W}\big)\,\bar{B}_k(X)\middle|X=x\right] = 0$$

for $\mathcal{P}_X$-almost every x and almost every c_2.

Expanding $\bar{B}_k(X)$:

$$\mathbb{E}_Z\big[\partial_2\mathcal{L}\big(Y, \hat{y}(X;\bar{W})\big)\big|X = x\big]\ \mathbb{E}_{C_2}\big[L_{C_2,k}\,\varphi_2'(\bar{H}_2(C_2; X, \bar{W}))\,\bar{w}_2(C_2)\big] = 0.$$

Step 5: Non-degeneracy

We note that our assumptions guarantee that $\mathbb{P}(\bar{w}_2(C_2) \neq 0)$ is positive. This is a crucial step that requires careful justification. In the full-width case, there are two cases:

- Case A: If $\int \mathbb{I}(u_2 \neq 0)\rho^2(du_2) > 0$ and $\xi_2(\cdot) = 0$ (second layer not trained), then it is obvious that $\mathbb{P}(\bar{w}_2(C_2) \neq 0) > 0$ since the initialization has non-zero mass.
- Case B: If $\xi_2(\cdot) \neq 0$ (second layer is trained) and $\mathscr{L}(w_1^0, w_2^0) < \mathbb{E}_Z[\mathcal{L}(Y, \varphi_2(0))]$, then by the gradient flow property (loss decreases along trajectories):

  $$\mathscr{L}(w_1^*(t), w_2^*(t)) \leq \mathscr{L}(w_1^*(t'), w_2^*(t')) \quad \text{for } t \geq t'.$$

  In particular, setting $t' = 0$ and taking $t \to \infty$:

  $$\mathscr{L}(\bar{w}_1, \bar{w}_2) \leq \mathscr{L}(w_1^0, w_2^0) < \mathbb{E}_Z[\mathcal{L}(Y, \varphi_2(0))].$$

  If $\mathbb{P}(\bar{w}_2(C_2) = 0) = 1$, then $\hat{y}(X; \bar{w}_1, \bar{w}_2) = \mathbb{E}_{C_2}[\bar{w}_2(C_2)\,\varphi_2(\bar{H}_2)] = 0$ (since $\bar{w}_2 = 0$), so $\mathscr{L}(\bar{w}_1, \bar{w}_2) = \mathbb{E}_Z[\mathcal{L}(Y, \varphi_2(0))]$, a contradiction.

Remark 3 (On the assumption $\mathscr{L}(w_1^0, w_2^0) < \mathbb{E}_Z[\mathcal{L}(Y, \varphi_2(0))]$. $]Thisassumptionmayseemrestrictive, butitisactual$

It requires that the initial loss is better than the trivial predictor $\hat{y} = 0$ (which corresponds to $w_2 = 0$).

For most reasonable initializations (e.g., small random weights), this condition is satisfied with high probability.

If this condition fails, then the network starts worse than the trivial solution, and convergence to a global minimizer (which must be at least as good as the trivial solution) is still possible, but the non-degeneracy argument needs to be modified.

In practice, this assumption can be verified by checking the initial loss value.

Since φ_2' is strictly non-zero and the mixing kernel L has non-zero entries, we have:

$$\mathbb{E}_Z\big[\partial_2\mathcal{L}\big(Y, \hat{y}(X;\bar{W})\big)\big|X = x\big] = 0$$

for $\mathcal{P}_X$-almost every x.

Step 6: Global optimality

Since $\mathcal{L}$ is convex in its second variable, for any measurable function $\tilde{y}(x)$:

$$\mathcal{L}\big(y, \tilde{y}(x)\big) - \mathcal{L}\big(y, \hat{y}(x;\bar{W})\big) \geq \partial_2\mathcal{L}\big(y, \hat{y}(x;\bar{W})\big)\big(\tilde{y}(x) - \hat{y}(x;\bar{W})\big).$$

Taking expectation:

$$\mathbb{E}_Z\big[\mathcal{L}\big(Y, \tilde{y}(X)\big)\big] \geq \mathscr{L}(\bar{w}_1, \bar{w}_2),$$

i.e., $(\bar{w}_1, \bar{w}_2)$ is a global minimizer.

Step 7: Connection to the limit (exact adaptation)

Finally, to connect $\mathscr{L}(W^*(t))$ with $\mathscr{L}(\bar{w}_1, \bar{w}_2)$ in the limit $t \to \infty$, we have (exactly as in the full-width case):

$$\begin{aligned}
\big|\mathscr{L}(W^*(t)) - \mathscr{L}(\bar{w}_1, \bar{w}_2)\big| &= \big|\mathbb{E}_Z\big[\mathcal{L}\big(Y, \hat{y}^*(t,X)\big) - \mathcal{L}\big(Y, \hat{y}(X;\bar{W})\big)\big]\big| \\
&\le K\,\mathbb{E}_Z\big[\big|\hat{y}^*(t,X) - \hat{y}(X;\bar{W})\big|\big] \\
&\le K\,\mathbb{E}_{\pi_t}\Big[|w_2^*(t,C_2') - \bar{w}_2(C_2)| \\
&\qquad + |\bar{w}_2(C_2)| \max_{1\le k\le r} |w_1^*(t,C_1',k) - \bar{w}_1(C_1,k)|\Big],
\end{aligned}$$

where the last step uses:

- $\hat{y}(X;W) = \mathbb{E}_{C_2}[w_2(C_2)\,\varphi_2(H_2(C_2;X,W))]$
- The Lipschitz property of φ_2 and the bi-Lipschitz bound for H_2 from Lemma 1
- The structure: $|\hat{y}^*(t) - \hat{y}(\bar{W})| \le K\mathbb{E}_{C_2}[|w_2^*(t) - \bar{w}_2| + |\bar{w}_2|\,|H_2^*(t) - \bar{H}_2|]$

By Assumption 4, the right-hand side converges to 0 as $t \to \infty$. This completes the proof.

6 Key Differences from Full-Width Case

1. **Simplification: No algebraic topology needed**: Since we do not train ρ^1 (the first layer uses frozen random features), we avoid the complex arguments needed to show that $\mathrm{Law}(w_1^*(t,C_1))$ has full support. Instead, we rely on the fact that $f_1(C_1)$ are frozen random features with full support, which directly gives the dense span property.

2. **Low-rank structure in H_2**: The pre-activation H_2 is a sum over r channels, introducing a factor $\|L\|_{\infty,1} \le rK$ in the bi-Lipschitz bound (Lemma 1).

3. **Convergence assumption**: Instead of bounding differences for a single w_1, we take the maximum over r channels: $\max_{1\le k\le r} |w_1^*(t,c_1',k) - \bar{w}_1(c_1,k)|$. This is the key adaptation: the convergence assumption becomes:

$$\int (1 + |\bar{w}_2|)|\bar{w}_2| \max_{1\le k\le r} |\bar{w}_1(c_1,k)|\,|w_1^*(t,c_1',k) - \bar{w}_1(c_1,k)|\,d\pi_t \to 0.$$

4. **Backpropagated signal**: The signal B_k involves the mixing kernel L, but the structure remains similar: $B_k = \mathbb{E}_{C_2}[L_{C_2,k}\,\varphi_2'(H_2)\,w_2]$. The bound becomes:

$$|B_k^*(t) - B_k(\bar{W})| \le K(1 + |\bar{w}_2|)\Big(|w_2^*(t) - \bar{w}_2| + |\bar{w}_2| \max_k |w_1^*(t) - \bar{w}_1|\Big).$$

5. **Universal approximation**: The dense span property still holds because we have r channels with frozen random features $f_1(C_1)$ having full support. The low-rank structure does not reduce the expressivity below what is needed for global convergence, as long as r is sufficiently large.

7 Discussion on Assumptions

7.1 The Non-Degeneracy Assumption

The assumption that $\mathscr{L}(w_1^0, w_2^0) < \mathbb{E}_Z[\mathcal{L}(Y, \varphi_2(0))]$ (initial loss better than trivial predictor) is used to ensure non-degeneracy of the limit point. While this may seem restrictive, it is actually quite natural:

- **Practical relevance**: For most reasonable initializations (e.g., small random weights, Xavier initialization), this condition is satisfied with high probability. The initial network typically performs better than the trivial solution $\hat{y} = 0$.
- **Verifiability**: This assumption can be easily checked by computing the initial loss value. If it fails, one can either:
  1. Re-initialize the network
  2. Use a different initialization scheme
  3. Modify the proof to handle the degenerate case (which requires additional technical work)
- **Alternative conditions**: If $\xi_2(\cdot) = 0$ (second layer frozen) and the initialization has non-zero mass, then non-degeneracy is automatic without this assumption.
- **Connection to full-width case**: In the full-width three-layer case, the analogous assumption is $\mathscr{L}(w_1^0, w_2^0, w_3^0) < \mathbb{E}_Z[\mathcal{L}(Y, \varphi_3(0))]$, which has the same interpretation.

7.2 The Convergence Assumption

The convergence assumption (Assumption 4) is the most technical requirement. It states that the mean-field dynamics converge in a Wasserstein-like sense, with weights that are properly coupled. This assumption is typically verified by:

- Showing that the loss decreases along trajectories (gradient flow property)
- Establishing compactness of the trajectory set
- Using LaSalle's invariance principle or similar stability arguments

The low-rank structure enters through the $\max_k$ over r channels, but the overall structure of the assumption remains similar to the full-width case.

8 Summary

The proof strategy follows the same path as the full-width case:

1. Show convergence of the backpropagated signal using the convergence assumption
2. Use coupling arguments to relate $W^*(t)$ to $\bar{W}$
3. Bound differences using the bi-Lipschitz property of H_2 (with low-rank factor rK)
4. Establish orthogonality using the dense span property
5. Prove non-degeneracy of the limit point

6. Conclude global optimality using convexity
7. Connect the limit using the convergence assumption

The key technical contribution is showing that the low-rank structure (factor r) does not break the convergence argument, as long as the universal approximation property is maintained (which is ensured by having r channels with full support).

9 Extension to L Layers

We now extend the mean-field framework to L-layer low-rank random feature networks. The structure generalizes naturally from the three-layer case.

9.1 Architecture for L Layers

Consider an L-layer network with the following structure:

- **Layer 1**: Random features $\varphi_1(\langle f_1(C_1), X\rangle)$ where $f_1(C_1)$ are frozen random vectors and C_1 indexes the first-layer channels. **This layer is not trained.**
- **Layers 2 to $L-1$**: Each layer $\ell \in \{2, \dots, L-1\}$ employs a low-rank mixing matrix $L^{(\ell)}$ of rank r, producing pre-activations

$$H_\ell(t, c_\ell; X, W) = \sum_{k=1}^{r} L^{(\ell)}_{c_\ell,k}\, m_k^{(\ell)}(t; X, W),$$

  where $m_k^{(\ell)}(t; X, W)$ are the r channel-wise partial functions at layer ℓ.
- **Layer L (output)**: Trainable weights $w_L(t, C_L)$ producing the final output.

9.2 Mean-Field ODE System for L Layers

The mean-field ODE system consists of $(L-2) \times r + 1$ trainable equations:

- For each intermediate layer $\ell \in \{2, \dots, L-1\}$: r equations for $w_\ell(t, c_\ell, k)$, $k = 1, \dots, r$
- For the output layer L: 1 equation for $w_L(t, c_L)$

Total: $(L-2) \times r + 1 \approx rL$ equations for large L (since layer 1 uses frozen random features and is not trained).

For example:

- $L = 3$: $r + 1$ equations (as in the three-layer case)
- $L = 4$: $2r + 1$ equations
- $L = 5$: $3r + 1$ equations
- General L: $(L-2)r + 1$ equations

For layer $\ell \in \{2, \dots, L-1\}$ and channel $k \in \{1, \dots, r\}$:

$$\partial_t w_\ell(t, c_\ell, k) = -\xi_\ell(t)\, \mathbb{E}_Z\Big[d_L\big(Z; W(t)\big)\, \varphi'_\ell\Big(H_\ell(t, c_\ell; X, W(t))\Big)\, B_k^{(\ell)}(t; X)\Big], \tag{6}$$

where $B_k^{(\ell)}(t; X)$ is the channel-k backpropagated signal at layer ℓ:

$$B_k^{(\ell)}(t; X) = \mathbb{E}_{C_{\ell+1}}\Big[L^{(\ell+1)}_{C_{\ell+1},k}\, \varphi'_{\ell+1}\Big(H_{\ell+1}(t, C_{\ell+1}; X, W(t))\Big)\, w_{\ell+1}(t, C_{\ell+1})\Big].$$

For the output layer L:

$$\partial_t w_L(t, c_L) = -\xi_L(t)\, \mathbb{E}_Z\Big[d_L\big(Z; W(t)\big)\, \varphi_L\Big(H_L(t, c_L; X, W(t))\Big)\Big], \tag{7}$$

where $H_L(t, c_L; X, W)$ is defined recursively from the previous layers.

9.3 Recursive Structure of Pre-Activations

For $\ell \in \{2, \dots, L\}$, the pre-activations satisfy the recursive relation:

$$H_\ell(t, c_\ell; X, W) = \sum_{k=1}^{r} L^{(\ell)}_{c_\ell,k}\, m_k^{(\ell)}(t; X, W),$$

where

$$m_k^{(\ell)}(t; X, W) = \mathbb{E}_{C_{\ell-1}}\Big[w_{\ell-1}(t, C_{\ell-1}, k)\, \varphi_{\ell-1}\Big(H_{\ell-1}(t, C_{\ell-1}; X, W(t))\Big)\Big].$$

The model output is:

$$\hat{y}(X; W(t)) = \mathbb{E}_{C_L}\big[w_L(t, C_L)\, \varphi_L\big(H_L(t, C_L; X, W(t))\big)\big].$$

9.4 Assumptions for L Layers

Assumption 5 (Initialization for L layers)**.** *For each layer $\ell \in \{2, \dots, L\}$, the initial functions satisfy:*

$$\sup_{m \ge 1} \frac{1}{\sqrt{m}} \max_{1 \le k \le r} \mathbb{E}_{C_\ell}\big[|w_\ell^0(C_\ell, k)|^m\big]^{1/m} \le K \quad \textit{for } \ell \in \{2, \dots, L-1\},$$
$$\sup_{m \ge 1} \frac{1}{\sqrt{m}} \mathbb{E}_{C_L}\big[|w_L^0(C_L)|^m\big]^{1/m} \le K.$$

Assumption 6 (Regularity for L layers)**.** *There exists $K \ge 1$ such that:*

- *For each $\ell \in \{2, \dots, L\}$, ξ_ℓ are bounded on compact intervals and piecewise continuous.*
- *For each $\ell \in \{2, \dots, L\}$, $\varphi_\ell, \varphi'_\ell$ are K-bounded and K-Lipschitz; moreover φ'_ℓ is non-vanishing for $\ell \in \{2, \dots, L-1\}$.*
- *For each $\ell \in \{2, \dots, L-1\}$, the mixing kernels $L^{(\ell)}$ satisfy $\sup_{c_\ell,k} |L^{(\ell)}_{c_\ell,k}| \le K$.*
- *All other regularity conditions from Assumption 3 hold.*

Assumption 7 (Convergence for L layers)**.** *There exist functions $\bar{w}_\ell$ for $\ell \in \{2, \dots, L\}$ such that as $t \to \infty$, there exists a coupling π_t such that for each intermediate layer $\ell \in \{2, \dots, L-1\}$:*

$$\int (1 + |\bar{w}_L|) \prod_{j=\ell+1}^{L-1} (1 + |\bar{w}_j|) \max_{1 \le k \le r} |\bar{w}_\ell(c_\ell, k)| \, |w_\ell^*(t, c'_\ell, k) - \bar{w}_\ell(c_\ell, k)| \, d\pi_t \to 0,$$

and for the output layer:

$$\int (1 + |\bar{w}_L|) \, |w_L^*(t, c'_L) - \bar{w}_L(c_L)| \, d\pi_t \to 0.$$

9.5 Key Technical Extensions

Lemma 5 (Bi-Lipschitz property for L layers)**.** *For any W' and W'' satisfying the regularity assumptions, and for any layer $\ell \in \{2, \dots, L\}$:*

$$\left| H_\ell(t, c_\ell; X, W') - H_\ell(t, c_\ell; X, W'') \right| \le K^L \, \|L^{(\ell)}\|_{\infty,1} \max_{2 \le j \le \ell} \max_{1 \le k \le r} \mathbb{E}_{C_j} \left[|w'_j(t, C_j, k) - w''_j(t, C_j, k)| \right],$$

where the constant K^L comes from the product of Lipschitz constants across layers.

Remark 4 (Scaling with depth)**.** *The key observation is that the low-rank structure at each layer introduces a factor of r (through $\|L^{(\ell)}\|_{\infty,1} \le rK$), but these factors multiply across layers. However, the well-posedness argument remains valid as long as the product r^L is controlled, or more precisely, as long as the ψ_2 bounds remain finite. In practice, for fixed r and moderate L, this is not an issue.*

9.6 Proof Strategy for L Layers

The proof strategy extends naturally from the three-layer case:

1. **Well-posedness**: The Picard iteration argument extends to L layers, with the key modification that differences propagate through $L-1$ layers. The contraction constant becomes C^L where C is the single-layer constant, but this is still manageable for fixed L.

2. **Global convergence**: The same argument applies, but now the backpropagated signal $B_k^{(\ell)}$ depends on all subsequent layers. The key bound becomes:

$$|B_k^{(\ell)*}(t) - B_k^{(\ell)}(\bar{W})| \le K^L (1 + |\bar{w}_L|) \prod_{j=\ell+1}^{L-1} (1 + |\bar{w}_j|) \Big(|w_L^*(t) - \bar{w}_L| + \sum_{j=\ell+1}^{L-1} \max_k |w_j^*(t) - \bar{w}_j| \Big).$$

3. **Convergence assumption**: The assumption must ensure that all layer-wise differences converge, with appropriate weights that account for the depth. The product structure $(1 + |\bar{w}_L|) \prod_{j=\ell+1}^{L-1} (1 + |\bar{w}_j|)$ ensures that the bound remains finite.

Remark 5 (Computational complexity)**.** *The number of mean-field equations scales as $(L-2) \times r + 1 \approx rL$ for large L, which is linear in both the depth L and the rank r. This is much more efficient than the full-width case, which would require $O(N^L)$ equations.*